

Practical – 1

Branch = CSE(AI&ML)

Enrollment no.=25SE02ml465

Code = SECE2291

Subject = Data Structure

Aim: Working with the array.

Q1. Write a program to read numbers and store it in array and display it?

Ans1. INPUT :

```
#include <stdio.h>
int
main()
{
    int arr[50],i,n;
    printf("Enter the
    number of elements:");
    scanf("%d",&n);
    printf("Enter the elements:");
    for(i=0;i<5;i++)
    {
        scanf("%d",&arr[i]);
    }
    printf("The array is :\n");
    for(i=0;i<5;i++)
    {
        printf(" %d ",arr[i]);
    }
    return 0;
}
```

Output :

```
PS C:\Users\prakh\DSA_code> .\a.exe
Enter the number of elements:5
Enter the elements:12
13
14
15
16
The array is :
 12 13 14 15 16
PS C:\Users\prakh\DSA_code> █
```

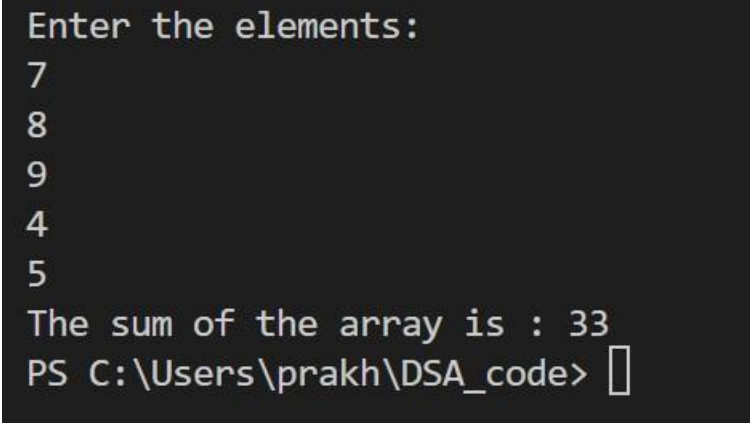
Q2. Write a program to demonstrate the concept of one-dimensional array finding the sum of array ?

Ans2. Input :

```
#include <stdio.h> int
main()
{
    int arr[50],i,n,sum=0;   printf("Enter the
number of elements:");   scanf("%d",&n);
printf("Enter the elements: \n");
for(i=0;i<n;i++)
{
    scanf("%d",&arr[i]);
}
for(i=0;i<n;i++)
{
    sum = sum + arr[i];
}
printf("The sum of the array is : %d",sum);
```

```
    return 0;
}
```

Output :



```
Enter the elements:
7
8
9
4
5
The sum of the array is : 33
PS C:\Users\prakh\DSA_code>
```

Q3. Write a program to insert an element in array ?

Ans 3. Input :

```
#include <stdio.h> int
main()
{
    int arr[50],i,n,pos, element ;
    printf("Enter the number of elements:");
    scanf("%d",&n);   printf("Enter the
elements: \n");   for(i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    printf("Enter the position where you want to insert an element:");
    scanf("%d",&pos);   printf("Enter the element to insert:");
    scanf("%d",&element);

    for(i=n-1;i>=pos;i--)
    {
        arr[i] =
arr[i-1];
```

```

    }

    arr[pos] = element;    n++;

    printf("The array after insertion is: \n");

    for(i=0;i<n;i++)

    {

        printf("%d ",arr[i]);

    }

    return 0;

}

```

Output :

```

PS C:\Users\prakh\DSA_code> gcc Array.pos.c -o Array
PS C:\Users\prakh\DSA_code> gcc Array.pos.c
PS C:\Users\prakh\DSA_code> .\a.exe
Enter the number of elements:5
Enter the elements:
7
8
9
9
4
Enter the position where you want to insert an element:3
Enter the element to insert:77
The array after insertion is:
7 8 9 77 9 14421096
PS C:\Users\prakh\DSA_code> 

```

Q4. Write a program to delete an element from an array ?

Ans4. Input :

```

#include <stdio.h> int

main()

{

```

```

    int arr[50],i,n,pos, element ;
printf("Enter the number of elements:");
scanf("%d",&n);  printf("Enter the
elements: \n");  for(i=0;i<n;i++)
{
    scanf("%d",&arr[i]);
}

    printf("Enter the position of the element you want to delete:");
scanf("%d",&pos);  if (pos<1 || pos>n)
{
    printf("invalid position");
return 0;
}

    for(i=pos-1;i<n-1;i++)
    {
        arr[i] = arr[i+1];
    }
n--;

    printf("The array after deletion is: \n");
for(i=0;i<n;i++)
{
    printf("%d ",arr[i]);
}

    return 0;
}

```

Output :

```

PS C:\Users\prakh\DSA_code> gcc Array.delete.c -o Array
PS C:\Users\prakh\DSA_code> gcc Array.delete.c
PS C:\Users\prakh\DSA_code> .\a.exe
Enter the number of elements:5
Enter the elements:
14
45
12
13
15
Enter the position of the element you want to delete:4
The array after deletion is:
14 45 12 15
PS C:\Users\prakh\DSA_code> 

```

Q5. Write a program to add two matrix A and B ?

Ans 5 . Input :

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int A[10][10], B[10][10], C[10][10];
```

```
    int rows, cols, i, j;
```

```
        printf("Enter number of rows and columns: ");
```

```
    scanf("%d %d", &rows, &cols);
```

```
        printf("Enter elements of Matrix A:\n");
```

```
        for(i = 0; i < rows; i++)
```

```
        {    for(j = 0; j < cols;
```

```
            j++)
```

```
            {
```

```
                scanf("%d", &A[i][j]);
```

```
}  
}
```

```
printf("Enter elements of Matrix B:\n");  
for(i = 0; i < rows; i++)  
{  
    for(j = 0; j < cols; j++)  
    {  
        scanf("%d", &B[i][j]);  
    }  
}  
for(i = 0; i < rows; i++)  
{  
    for(j = 0; j < cols; j++)  
    {  
        C[i][j] = A[i][j] + B[i][j];  
    }  
}
```

```
printf("Sum of Matrix A and Matrix B:\n");  
for(i = 0; i < rows; i++)  
{  
    for(j = 0; j < cols; j++)  
    {  
        printf("%d ", C[i][j]);  
    }  
    printf("\n");  
}
```

```
return 0;
```

```
}
```

Output :

```
PS C:\Users\prakh\DSA_code> .\a.exe
Enter number of rows and columns: 3
3
Enter elements of Matrix A:
1
2
2
3
4
5
6
7
8
Enter elements of Matrix B:
9
10
11
12
13
14
15
16
17
Sum of Matrix A and Matrix B:
10 12 13
15 17 19
21 23 25
```

Q6. Write a program to concatenate two strings ?

Ans 6. Input :

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
    char str1[100], str2[100];
```

```
printf("Enter first string: ");  
scanf("%s", str1);  
  
printf("Enter second string: ");  
scanf("%s", str2);  
  
strcat(str1, str2);  
  
printf("Concatenated String: %s\n", str1);  
  
return 0;  
}
```

Output :

```
PS C:\Users\prakh\DSA_code> gcc Array.conc.c -o Array  
PS C:\Users\prakh\DSA_code> gcc Array.conc.c  
PS C:\Users\prakh\DSA_code> .\a.exe  
Enter first string: Prakh  
Enter second string: Muraliya  
Concatenated String: PrakhMuraliya  
PS C:\Users\prakh\DSA_code> 
```